

Birla Institute of Technology and Science, Pilani

Work Integrated Learning Programmes Division

M.Tech. in Artificial Intelligence and Machine Learning.

I Semester 2025-26

Course Number	AIMLZC416	
Course Name	Mathematical Foundations for Machine Learning	
Nature of Exam	Closed Book	# Pages 3
Weightage for grading	30%	# Questions 7
Duration	120 minutes	
Date of Exam	20/09/2025 (FN: 09:00 - 11:00)	

Instructions

1. All questions are compulsory
 2. Marks are indicated against each question
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Q1. We are given that the $p \times n$ matrix V can be written as the product SA where S is a $p \times m$ matrix and A is a $m \times n$ matrix, where $n > m$. We claim that the columns of V are linearly independent. Can you ascertain if this claim is true or false, or do we need more information before we can make this claim? Give detailed justification for your answer. (4 marks)

Q2. Consider the intersection of two hyperplanes $a^T x = \alpha$ and $b^T x = \beta$, where $b \neq ka$ for any value k and x is a $n \times 1$ vector where $n > 2$. Do there exist values of α and β for which the intersection of these two hyperplanes forms a vector space? If so, for what values of α and β and what is the dimension of the vector space so formed? Provide a detailed justification for your answer, regardless of whether or not a vector space can be formed. (4 marks)

Q3.(a) Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^{n \times 1}$ be two vectors. Consider the identity matrix $\mathbf{I}_n \in \mathbb{R}^{n \times n}$.

- (i) Consider the matrix $\mathbf{B} = \mathbf{I}_n - 2(\mathbf{v}\mathbf{v}^T)$. Prove or disprove whether this matrix $\mathbf{B}$ is an orthogonal matrix. Also prove/disprove whether $\mathbf{B}$ is a symmetric matrix. (1.5 marks)
- (ii) Assume $(\|\mathbf{w}\|_2)^2 = 1$. Consider the matrix $\mathbf{C} = \mathbf{I}_n - 2(\mathbf{w}\mathbf{w}^T)$. Prove or disprove whether this matrix $\mathbf{C}$ is an orthogonal matrix. (1.5 marks)

(b) Consider the symmetric positive definite matrix $\mathbf{A}$ given below:

$$\mathbf{A} = \begin{bmatrix} 7 & -1 \\ -1 & 7 \end{bmatrix}$$

and an inner product defined as $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{A} \mathbf{y}$. Find the angle between vectors $\mathbf{a} = [2 \ 1]^T$ and $\mathbf{b} = [1 \ 3]^T$ in the above defined inner product space. (1 mark)

Q4.(a) A professor teaching Linear Algebra told students to consider a matrix $\mathbf{B} \in \mathbb{R}^{n \times n}$. He also told students that $\mathbf{B}$ is an orthogonal matrix. A student named M in class then claimed that the determinant of $\mathbf{B}$ can take only two possible values due to the orthogonality property of the matrix $\mathbf{B}$. Do you think the claim made by M is true ?. Give proper reasoning for your answer. If yes,

provide a proof. If not, provide a counter example. (1 mark)

(b) Consider the positive definite matrix $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ defined as:

$$\mathbf{A} = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}$$

- (i) Derive a lower triangular matrix $\mathbf{L}$ so that $\mathbf{A}$ can be factorized as $\mathbf{A} = \mathbf{L}\mathbf{L}^T$. Derive all the three eigenvalues of $\mathbf{L}$. (1.5 marks)
- (ii) Let $\mathbf{x} \in \mathbb{R}^n$ be a nonzero vector. Define $\mathbf{z} = \mathbf{L}^T \mathbf{x}$ where $\mathbf{L} \in \mathbb{R}^{3 \times 3}$ satisfies the equality $\mathbf{A} = \mathbf{L}\mathbf{L}^T$. Prove that $\mathbf{z}$ can never be the zero vector. (1.5 marks)

Q5. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function defined by

$$f(x, y) = e^x \cos(y).$$

- (a) Derive the second-order Taylor expansion of $f(x, y)$ around $(0, 0)$:

$$f(x, y) \approx f(0, 0) + \nabla f(0, 0) \cdot (x, y) + \frac{1}{2} (x, y) H_f(0, 0) (x, y)^T.$$

(1 mark)

- (b) Use this to estimate $f(1, \frac{\pi}{4})$, where $H_f(x, y)$ is the Hessian matrix of $f(x, y)$. (1 mark)
- (c) Using the exact value of $f(1, \frac{\pi}{4})$, compute the approximation error of the Taylor expansion. (1 mark)
- (d) Discuss how the Hessian matrix helps determine whether the quadratic approximation overestimates or underestimates the true value of f near $(0, 0)$. (1 mark)

Q6. Consider the function $f(\bar{X}) = \sigma(W^T \bar{X} + \bar{b})$, where $\bar{X} \in \mathbb{R}^n$ is the input vector, $W \in \mathbb{R}^{m \times n}$ is the weight matrix, $\bar{b} \in \mathbb{R}^m$ is the bias vector, and $\sigma(\cdot)$ is the element-wise sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- (a) Draw the computational graph for $f(\bar{X})$, explicitly showing intermediate nodes for $z = W^T \bar{X} + \bar{b}$ and $\sigma(z)$. (1 mark)
- (b) For a given input $\bar{X} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, weight matrix $W = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}$, $\bar{b} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, compute the forward propagation output $\mathbf{y} = \mathbf{f}(\bar{\mathbf{X}})$. (2 marks)
- (c) Compute the loss function $L = \frac{1}{2} \|\mathbf{y} - \mathbf{t}\|^2$, where $\mathbf{t} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is the target vector. (1 mark)

Q7.(a) Find the SVD from the eigenvectors v_1, v_2 of $A^T A$ and $Av_i = \sigma_i u_i$:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad (2 \text{ marks})$$

(b) Given a matrix $A_{m \times n}$ with $m \neq n$, write an algorithm / pseudocode to determine the singular value decomposition, using only the function $[E, V] =$

$\text{eigs}(C)$ which gives the eigenvectors of C in E and eigenvalues in the diagonal matrix V . (2 marks)

(c) For a real symmetric positive semi-definite matrix, how are its singular values related to its eigenvalues? (2 marks)